key Modules

A) Directory (Annuaire)

The directory is essentially **search + browse** over profiles.

Directory requirements

1. **Search**
   - Full-text search on name/bio/skills (and maybe organization)
2. **Filters**
   - Location, skills/tags, role, organization
3. **Sorting**
   - Alphabetical, newest, most active (optional)
4. **Pagination**
   - Mandatory for performance
5. **Access control**
   - Public directory or members-only directory (choose policy; backend must support both)

Concrete backend responsibilities

- Provide list endpoints optimized with pagination & indexes.
- Return only fields allowed by privacy settings.

B) Profiles (Profil par personne)

A profile is a structured object connected to a user (or standalone directory person).

Profile requirements (refined)

1. **Profile fields**
   - Required: display name
   - Suggested: photo URL, bio, location, roles/skills, organization, links (LinkedIn/GitHub), interests
   - Contact fields: email/phone should be **privacy-controlled**
2. **Privacy controls**
   - Per-field visibility: `public`, `members_only`, `private`
   - Global profile visibility: `public` / `members_only` / `hidden`
3. **Profile completeness**
   - Backend computes completeness score (optional) to encourage filling info
4. **Moderation workflow (optional)**
   - `draft -> pending_review -> published -> suspended`

Concrete backend responsibilities

- Validate and store profile updates.

- Enforce privacy rules when returning profile data.
- Provide search/filter over allowed fields.

C) Events (Événements)

Event requirements (refined)

1. **Event CRUD**
   - Create event: title, description, start/end datetime, location (online/offline), capacity, organizer, visibility
   - Update / cancel event (status changes)
2. **Visibility**
   - `public`, `members_only`, `private_invite`
3. **Registrations / RSVP**
   - Member can RSVP: `going`, `interested`, `not_going`
   - Track capacity: waitlist if full (optional)
4. **Event communication**
   - Send email to attendees (admin/organizer)
   - Reminders (e.g., 24h before) (optional but common)
5. **Auditability**
   - Keep who created/updated/cancelled an event

Concrete backend responsibilities

- Prevent overbooking if capacity is set.
- Ensure RSVP operations are idempotent (same request doesn't create duplicates).
- Enforce visibility rules for event listing and details.

D) Email subscriptions (emails pour abonnements)

This is separate from authentication email (verification/reset).

Subscription requirements

1. **Subscription topics**
   - At minimum:
     - `newsletter`
     - `event_announcements`
     - `event_reminders`
   - Later: forum notifications, digests, mentions
2. **Double opt-in (recommended)**
   - user subscribes → receives confirmation email → confirmed
3. **Unsubscribe**
   - One-click unsubscribe link with secure token
   - Per-topic unsubscribe

4. **Compliance**
    - Store consent timestamp + source (where they subscribed)
5. **Sending**
    - Queue-based sending (don’t send synchronously in request threads)
    - Track status: sent, bounced, complaint (depends on provider)

Concrete backend responsibilities

- Maintain subscription states reliably.
- Generate secure tokens for unsubscribe/confirm flows.
- Provide admin endpoints to trigger campaigns (MVP can be simple).

E) Identity & Access (Inscription + Login)

Features

1. **Register account**
    - Input: email, password, name (minimum)
    - Store hashed password
    - Email verification (recommended)
    - Rate limiting & anti-abuse controls
2. **Login**
    - Input: email + password
    - Output: session cookie or JWT access token (+ refresh token)
3. **Password reset**
    - Request reset email
    - Apply reset token
4. **Account lifecycle**
    - Deactivate account
    - Delete account (GDPR-like) + anonymize related content (important for forum later)

Acceptance criteria

- Duplicate email is rejected.
- Password policies enforced (length, complexity optional).
- Email verification required before being visible in the directory (optional but common).